# A Self-Paced, Proof-Level Syllabus

Reading roadmap to fully understand — and explain —

*"The Noisy Folded Limit Cycle: Tracy–Widom Statistics of Canard Escape"*

**The paper in one paragraph**

A slow–fast oscillator carries a one-parameter family of limit cycles that *folds*: an attracting and a saddle cycle collide and annihilate at a *fold of limit cycles*. Near that fold a geometric blow-up (Krupa–Szmolyan, adapted to cycles by Jelbart–Kuehn–Kuntz) reduces the amplitude dynamics to the canonical *noisy Riccati* equation $\mathrm{d}R = (R^2 - Y)\,\mathrm{d}T + \eta\,\mathrm{d}B$ with $Y = Y_0 - T$ and effective noise $\eta = \sigma/\sqrt{\varepsilon_2}$. Deterministically this is the Airy equation, whose canard solution $R = \mathrm{Ai}'(Y)/\mathrm{Ai}(Y)$ escapes at the first Airy zero $a_1 \approx -2.3381$. The substitution $R = -u'/u$ (Cole–Hopf) linearises the noisy equation to a *Schrödinger equation with a white-noise potential*, $u'' = (Y - \eta\xi)u$ — exactly the stochastic Airy operator $\mathcal{H}_\beta = -\partial_x^2 + x + \frac{2}{\sqrt{\beta}}b'$ of Ramírez–Rider–Virág, with dictionary $\eta = 2/\sqrt{\beta}$. Its ground-state eigenvalue is Tracy–Widom distributed, so the slow-variable value at canard escape ("peel-off") obeys $Y_{\text{node}} \overset{d}{=} -\Lambda_0(\beta) \overset{d}{=} \mathrm{TW}_\beta$, $\beta = 4/\eta^2$. Consequences: early-escape probability $1 - F_\beta(0)$; a large-deviation rate constant $c = 5.4439\ldots$ that is a Painlevé II action; a physical critical-noise law $\sigma_* \propto \sqrt{\varepsilon_2}$; and, in the phase channel when the cycle dies at a saddle-node-on-invariant-circle, a bifurcation-specific 2/3 exponent governed by the Adler / noisy-saddle-node first-passage law. The amplitude channel places canard escape in the Tracy–Widom / KPZ edge universality class.

To read every line of that paragraph as a theorem with a proof you need four bodies of theory — geometric singular perturbation theory, stochastic analysis, random matrix theory, and the special-function/spectral toolkit — resting on a foundation of measure-theoretic probability and operator theory. This syllabus builds them in dependency order. It assumes you are comfortable with undergraduate analysis and linear algebra and that you already know basic dynamical systems and bifurcations (FitzHugh–Nagumo, saddle-node, Hopf). Everything past that is constructed here, to proof level.

## 1 How to use this syllabus

The roadmap has three layers. **Foundations (F1–F4)** are gate modules: shore these up first, in any order, though F2 should precede the probability-heavy tracks. The **four pillars (A–E)** are the substantive tracks; each ends by "unlocking" specific equations, lemmas, and sections of the paper. The **capstone (S1–S2)** assembles the pillars into the paper's hardest original step and then a full read-through.

Each module follows the same template: a *Goal*; *prerequisites* (by module label); a list of *core topics* stated at the level of rigour you should reach; *primary reading* with specific chapters; *going deeper*; *milestone problems* (do these — they are how you convert reading into the ability to reproduce the paper); and a box stating exactly *what it unlocks in the paper*.

Two passes are recommended even though the target is proof-level. First a fast pass for the narrative (read each module's Goal and "unlocks" box, skim the primary text's introductions). Then the rigorous pass, working the milestone problems. Full bibliographic details and links are in Section 9; a one-page checklist is in Section 10; and Section 11 lists the questions you should be able to answer cold once you are done — that is the "explain it to someone" deliverable.

**Dependency map**

| Module | Builds on | Unlocks (paper) |
|---|---|---|
| F1 ODEs & bifurcations | — | §2.1 normal form; $\alpha$-dichotomy |
| F2 Measure-theoretic probability | — | footing for B, C |
| F3 Functional analysis & Sturm–Liouville | F2 | §4.2 Sturm fact; §4.3 monotonicity |
| F4 Airy, Riccati, Cole–Hopf | F1 | §2.2 canard; Lemma 6 |
| A GSPT & geometric blow-up | F1, F4 | all of §2; §6 |
| B Stochastic calculus, Berglund–Gentz, large deviations | F2, (A) | Lemma 6; §5; §6 tubes; §7; Prop. 8 |
| C Random matrices & the stochastic Airy operator | F2, F3, F4, B | §3; §4.2; §8; §9 |
| D Painlevé II & the exact laws | F4, C | §3 Thm 3; §5 constant $c$ |
| E Excitable systems & the phase channel | F1, B | §7; §9.2 |
| S1 Shooting characterisation (keystone) | A, B, C, F3 | §4.3 |
| S2 Full read-through | all | whole paper |

# 2 Foundations

## 2.1 Module F1 — Rigorous ODEs, invariant manifolds, and bifurcation theory

**Goal**

Upgrade "I know what a saddle-node and a Hopf are" to "I can state and prove the normal-form and centre-manifold theorems, and I recognise a fold of limit cycles."

*Prerequisites:* undergraduate ODEs and linear algebra.

**Core topics.**

- Existence/uniqueness, flows, linearisation, hyperbolicity; stable/unstable/centre manifolds and the centre-manifold reduction (with proof sketch).
- Local bifurcations of equilibria: saddle-node and Andronov–Hopf normal forms; topological vs. smooth normal form; universal unfoldings.
- Bifurcations of *periodic orbits*: the Poincaré/return map, Floquet multipliers, the saddle-node (fold) of limit cycles, and the homoclinic/SNIC routes by which a cycle can die. This is the deterministic object the paper perturbs.
- Averaging and the phase description of oscillators (sets up the $\theta$ variable and the $\alpha$-dichotomy).

**Primary reading.** Kuznetsov, *Elements of Applied Bifurcation Theory*, ch. 2–5 (equilibria, centre manifolds, codim-1 normal forms) and the periodic-orbit bifurcation material. Strogatz, *Nonlinear Dynamics and Chaos*, for a fast geometric refresher (esp. relaxation oscillators and the Adler/overdamped-pendulum equation).

**Going deeper.** Guckenheimer & Holmes, *Nonlinear Oscillations...*, ch. 3–4 (averaging, normal forms).

**Milestone problems.**

1. Prove the saddle-node normal form $\dot{x} = \mu \pm x^2$ via centre-manifold reduction for a 1-parameter planar family.
2. For a limit cycle, define the Poincaré map and show a fold of cycles is a saddle-node of its fixed point; identify the Floquet multiplier crossing $+1$.
3. Write the deterministic part of the paper's normal form (2) and explain why $a, b, c$ are positive 1-periodic functions of $\theta$.

**What this unlocks in the paper**

The deterministic skeleton of §2.1, the meaning of "fold of limit cycles," and (with E) the $\alpha = 1$ vs. $\alpha = 2$ phase dichotomy of §6.

## 2.2 Module F2 — Measure-theoretic probability

**Goal**

Reach the level where "$\overset{d}{=}$," "a.s.," "stationary white noise," and "a continuous functional of the Brownian path" are precise statements you can manipulate. This is the gate to Tracks B and C.

*Prerequisites:* real analysis (measure/integration helpful but built here).

**Core topics.**

- Probability spaces, random variables, expectation as Lebesgue integral, modes of convergence; laws of large numbers and the central limit theorem.
- Conditional expectation; filtrations; discrete-time martingales and the optional-stopping and convergence theorems.
- Weak convergence of measures, characteristic functions, and the meaning of convergence in distribution (the relation "$\overset{d}{=}$").
- First construction of Brownian motion (Wiener measure) — enough to state its scaling, stationarity, and reflection symmetry, which the paper uses crucially.

**Primary reading.** Durrett, *Probability: Theory and Examples*, ch. 1–5 [free PDF online](author posts a PDF). Williams, *Probability with Martingales*, for an exceptionally clean martingale treatment.

**Milestone problems.**

1. Prove convergence in distribution is equivalent to pointwise convergence of characteristic functions (Lévy continuity).
2. Show Brownian motion is stationary-increment and invariant under $t \mapsto -t$ reflection; state white noise $\xi = \mathrm{d}B/\mathrm{d}t$ as its (distributional) derivative. (Used verbatim in §4.3.)

**What this unlocks in the paper**

Every "$\overset{d}{=}$" in the theorems; the stationarity/reflection facts that make the random level in §4.3 collapse to a deterministic one; and the probabilistic meaning of the LDP in §5.

### 2.3 Module F3 — Functional analysis, self-adjoint operators, and Sturm–Liouville/oscillation theory

**Goal**

Be able to treat $-\partial_x^2 + V$ as a self-adjoint operator, define its spectrum and ground state, and prove the classical Sturm fact: *the first zero of the recessive solution, swept in the spectral parameter, is minus the ground-state energy.*

*Prerequisites:* F2-level analysis; Hilbert spaces.

**Core topics.**

- Hilbert spaces, bounded/unbounded operators, adjoints, symmetric vs. self-adjoint, the spectral theorem; quadratic forms and the Rayleigh–Ritz variational characterisation of the lowest eigenvalue.
- Sturm–Liouville theory and Weyl's limit-point/limit-circle dichotomy; recessive (principal) vs. dominant solutions at infinity.
- Sturm oscillation theory: nodes of eigenfunctions, comparison theorems, and **Dirichlet domain monotonicity** — shrinking the domain raises the ground state. This is the engine of the shooting argument (§4.3).

**Primary reading.** Teschl, *Mathematical Methods in Quantum Mechanics*, Part 1 (functional analysis, self-adjointness, spectral theorem) and the Sturm–Liouville/oscillation chapters [free PDF online].

**Going deeper.** Reed & Simon, *Methods of Modern Mathematical Physics I*, for the spectral theorem in full.

**Milestone problems.**

1. Prove the variational (min–max) characterisation of the lowest Dirichlet eigenvalue of $-\partial_x^2+V$ on a half-line.
2. Prove Dirichlet domain monotonicity: if $\Omega_1 \subset \Omega_2$ then $\lambda_0(\Omega_1) \geq \lambda_0(\Omega_2)$.
3. Prove the Sturm fact $Y_{\text{node}} = a_1 = -\Lambda_0(\infty)$ for the deterministic Airy operator: the first zero of the recessive solution equals minus the ground-state energy.

**What this unlocks in the paper**

The phrase in §4.2 "the recessive solution's first zero is minus the ground-state energy," and the monotone spectral function $G(\ell)$ that is the crux of Theorem 2 (§4.3).

### 2.4 Module F4 — The canard's special functions: Airy, Riccati, Cole–Hopf, and asymptotics

**Goal**

Know the Airy equation and its asymptotics cold; be able to pass between a Riccati equation and the linear second-order equation behind it; and master Watson's lemma / Laplace's method.

*Prerequisites:* F1; complex analysis basics.

**Core topics.**

- Airy functions $\mathrm{Ai}, \mathrm{Bi}$: defining ODE $u'' = xu$, integral representations, zeros (the first at $a_1 \approx -2.3381$), and asymptotics.

- The Riccati equation $R' = R^2 - Y$ and its linearisation $R = -u'/u \Rightarrow u'' = Yu$; a finite-time blow-up of $R$ is exactly a simple zero (node) of $u$. This is the deterministic Cole–Hopf identity behind Lemma 6.

- Asymptotic methods: Laplace's method, Watson's lemma, stationary phase, and the WKB and turning-point picture near a fold (where the Airy equation is the universal local model).

**Primary reading.** NIST *Digital Library of Mathematical Functions*, ch. 9 (Airy) and ch. 32 (Painlevé) [free PDF online]. Bender & Orszag, *Advanced Mathematical Methods*, ch. 3, 6 (asymptotics, WKB, Airy).

**Milestone problems.**

1. Show $R = \mathrm{Ai}'(Y)/\mathrm{Ai}(Y)$ solves $R' = R^2 - Y$ and blows up exactly at zeros of Ai; locate the first escape at $a_1 \approx -2.3381$.

2. Prove the deterministic Cole–Hopf statement: $R = -u'/u$ turns $R' = R^2 - Y$ into $u'' = Yu$, and a blow-up of $R$ is a node of $u$.

3. Use Watson's lemma to evaluate the leading asymptotics of $\int_0^\infty \sqrt{Y}\, e^{-8Y^{3/2}/3\eta^2}\, \mathrm{d}Y$ (the hazard integral, eq. (5)).

**What this unlocks in the paper**

The deterministic inner solution and escape point of §2.2; the algebraic skeleton of Lemma 6; the hazard integral (5) and $\eta_* = 2\sqrt{\pi}$; and Watson's lemma as used in the phase channel (§7).

# 3 Track A — Geometric singular perturbation theory and the blow-up

## 3.1 Module A — Fenichel theory, fold/canard points, and geometric desingularisation

**Goal**

Understand the paper's entire §2: critical manifolds and normal hyperbolicity, loss of hyperbolicity at a fold, the Krupa–Szmolyan blow-up with weights $(1, 2, 3)$, the charts $K_1, K_2, K_3$, and the Jelbart–Kuehn–Kuntz extension to a *manifold of limit cycles*.

*Prerequisites:* F1, F4.

**Core topics.**

- Fast–slow systems, critical manifold, Fenichel's invariant-manifold theorems and normal hyperbolicity; why hyperbolicity fails at a fold.

- Canards, relaxation oscillations, and mixed-mode oscillations; the folded node as the prototype of the "fold family."

- Geometric desingularisation (blow-up): the quasi-homogeneous map, weight choice (here $(1, 2, 3)$ on $(r, y, \varepsilon_2)$), and the entry/rescaling/exit charts $K_1, K_2, K_3$; how the rescaling chart $K_2$ produces the inner equation and the scalings $r = \varepsilon_2^{1/3} R$, $y = \varepsilon_2^{2/3} Y$, $T = \varepsilon_2^{1/3} t$.

- Folds of *limit cycles*: the JKK normal form and chart-by-chart passage — the deterministic backbone the paper adds noise to.

**Primary reading.** Kuehn, *Multiple Time Scale Dynamics*: ch. 1–3 (GSPT/Fenichel), ch. 7 (blow-up), ch. 8 (canards), ch. 13 (oscillations/MMOs). Jardón-Kojakhmetov & Kuehn, *A survey* *on the blow-up method for fast-slow systems*, arXiv:1901.01402 [free PDF online]— the fold case worked in detail.

**Going deeper.** Krupa & Szmolyan (2001), the original fold/canard blow-up (ref. [1]). Desroches *et al.*, *Mixed-Mode Oscillations with Multiple Time Scales*, SIAM Review (2012) [free PDF online]. Jelbart, Kuehn & Kuntz, *Geometric Blow-Up for Folded Limit Cycle Manifolds in Three Time-Scale Systems*, arXiv:2208.01361 / J. Nonlinear Sci. (2024) (ref. [2]).

**Milestone problems.**

1. Apply the $K_2$ rescaling (3) to normal form (2) and recover the inner Riccati (1); read off $\eta = \sigma/\sqrt{\varepsilon_2}$.
2. Derive the $(1, 2, 3)$ weights from the requirement that the fold normal form be quasi-homogeneously invariant; identify what each chart $K_1, K_2, K_3$ resolves.
3. In the deterministic inner equation, identify the canard $\mathrm{Ai}'/\mathrm{Ai}$ and the escape at the first Airy zero.

**What this unlocks in the paper**

All of §2 (normal form, blow-up, charts, the inner Riccati, $\eta = \sigma/\sqrt{\varepsilon_2}$) and the chart composition $\Pi_1 \circ \kappa_{12} \circ \Pi_2 \circ \kappa_{23} \circ \Pi_3$ of §6.

# 4 Track B — Stochastic calculus, sample-path tubes, and large deviations

## 4.1 Module B1 — Brownian motion, the Itô integral, and SDEs

**Goal**

Be able to compute with Itô's formula, know exactly when Itô and Stratonovich agree, and prove the paper's Lemma 6 (the Cole–Hopf identity with *no* Itô correction).

*Prerequisites:* F2, F4.

**Core topics.**

- Brownian motion, quadratic variation, the Itô integral, Itô's formula; the Itô–Stratonovich correction and why additive (constant-coefficient) noise has none.
- Existence/uniqueness for SDEs; diffusions, generators, the Fokker–Planck/Kolmogorov equations.
- Linear SDEs and stochastic flows; transforming SDEs under diffeomorphisms (the Itô computation behind Proposition 8).

**Primary reading.** Evans, *An Introduction to Stochastic Differential Equations* [free PDF online]— the gentlest rigorous entry. Øksendal, *Stochastic Differential Equations*, ch. 3–5 (Itô integral, formula, SDEs).

**Going deeper.** Karatzas & Shreve, *Brownian Motion and Stochastic Calculus*, ch. 2–5, for the fully rigorous treatment.

**Milestone problems.**

1. **(Lemma 6.)** For $\mathrm{d}u = v\,\mathrm{d}T,\ \mathrm{d}v = (Yu)\,\mathrm{d}T - \eta u\,\mathrm{d}B$ and $R = -v/u$, apply Itô's formula and show $\mathrm{d}R = (R^2 - Y)\,\mathrm{d}T + \eta\,\mathrm{d}B$ with no correction (the diffusion vector $g = (0, -\eta u)$ has $\frac{1}{2}(g \cdot \nabla)g = 0$ and $R_{vv} = 0$).
2. Prove the change-of-variables in Proposition 8: under (3) the physical SDE (2) becomes the inner Riccati with $\eta = \sigma/\sqrt{\varepsilon_2}$ exactly.

3. Derive the noise-induced peel-off scalings $r_{\text{peel}} \sim \sigma^{2/3}$, $y_{\text{peel}} \sim \sigma^{4/3}$.

**What this unlocks in the paper**

Lemma 6 and its "no Itô–Stratonovich correction" claim; Proposition 8 and the blow-down scalings of §6.

### 4.2 Module B2 — Berglund–Gentz sample-path theory

**Goal**

Learn the "tube" technology: how a stochastic slow–fast trajectory concentrates around the deterministic one, with Gaussian fluctuations in the hyperbolic regions and a different law near the fold.

*Prerequisites:* B1, A.

**Core topics.**

- Concentration of sample paths in neighbourhoods of slow manifolds; first-exit times and the size of fluctuation tubes.
- Slow passage through a fold/bifurcation with noise; the Kramers/mean-first-passage hazard and how a single separatrix crossing differs from a full escape.
- Error propagation through a chain of transition maps (the additive telescoping used in §6.1) and the role of exponential contraction at the fold.

**Primary reading.** Berglund & Gentz, *Noise-Induced Phenomena in Slow–Fast Dynamical Systems: A Sample-Paths Approach* (ref. [4]) — ch. on stable slow manifolds and on the fold.

**Going deeper.** Berglund, Gentz & Kuehn, *Hunting French ducks in a noisy environment* and *From random Poincaré maps to stochastic MMO patterns* (ref. [5]).

**Milestone problems.**

1. Reproduce the integrated Kramers hazard (5), $H(\eta) = \frac{1}{\pi}\int_0^\infty \sqrt{Y}\, e^{-8Y^{3/2}/3\eta^2}\, \mathrm{d}Y = \eta^2/4\pi$, and obtain $\eta_* = 2\sqrt{\pi}$.
2. Set up the transverse-error recursion $|\delta_i| \le L_i|\delta_{i-1}| + |\zeta_i|$ and telescope it; explain why exponential contraction at the fold annihilates the entry fluctuation.

**What this unlocks in the paper**

The hazard (5) and critical-noise scale; the "tubes" supplying the sub-dominant Gaussian corrections in charts $K_1, K_3$; and the composition/propagation estimates of §6.1 (Lemma 7).

### 4.3 Module B3 — Large deviations: Schilder, contraction, Freidlin–Wentzell

**Goal**

Be able to run the small-noise large-deviation argument that produces the rate constant $c$: write the ground-state eigenvalue as a continuous functional of the Brownian path and push Schilder's theorem through the contraction principle.

*Prerequisites:* F2, B1.

**Core topics.**

- Rate functions, the large-deviation principle, Varadhan's lemma; the contraction principle.

- Schilder's theorem (LDP for Brownian motion) and the Freidlin–Wentzell theory of small-noise diffusions; the quasipotential.

**Primary reading.** Dembo & Zeitouni, *Large Deviations Techniques and Applications*, ch. 4–5 (Schilder, contraction, Freidlin–Wentzell) (ref. [8]). Freidlin & Wentzell, *Random Perturbations of Dynamical Systems*, ch. 3–4.

**Milestone problems.**

1. Write $\Lambda_0(\eta) = \inf_{\|\psi\|=1,\psi(0)=0} (K[\psi] + \eta N[\psi])$ with $K[\psi] = \int(\psi'^2 + x\psi^2)$, $N[\psi] = \int \psi^2 \, \mathrm{d}W$; integrate by parts to make $N[\psi] = -2\int \psi\psi' W$ and exhibit $\Lambda_0$ as a continuous functional $F(\eta W)$ of the path.
2. Apply Schilder + contraction to get rate $\inf\{\frac{1}{2}\int g'^2 : F(g) \le 0\} = \frac{1}{2}\min_\psi K[\psi]^2/\int \psi^4 = c$; derive the Euler–Lagrange equation $-g'' + xg = g^3$.

**What this unlocks in the paper**

The entire rate-constant proof in §5: the LDP yielding $P_{\text{early}} = \exp(-c/\eta^2)$ and the variational problem whose minimiser is the nonlinear-Airy ground state.

# 5 Track C — Random matrices and the stochastic Airy operator

## 5.1 Module C — From beta ensembles to $\mathrm{TW}_\beta$ and the operator $\mathcal{H}_\beta$

**Goal**

Understand the spectral edge of random matrices: the Tracy–Widom laws $\mathrm{TW}_{1,2,4}$ and their extension to all $\beta > 0$, the Airy kernel, and — the keystone import — the Ramírez–Rider–Virág stochastic Airy operator $\mathcal{H}_\beta$ whose ground state *is* the inner Riccati of the paper.

*Prerequisites:* F2, F3, F4, B1.

**Core topics.**

- Gaussian ensembles (GOE/GUE/GSE), the Wigner semicircle law, and the bulk/edge distinction; $\beta = 1, 2, 4$ symmetry classes.
- Edge scaling, the Airy kernel, the Fredholm-determinant form of $F_2$, and the Tracy–Widom distributions; tail behaviour and skewness.
- Tridiagonal ($\beta$-ensemble) models; the continuum limit to the **stochastic Airy operator** $\mathcal{H}_\beta = -\partial_x^2 + x + \frac{2}{\sqrt{\beta}} b'$ on $[0, \infty)$ with Dirichlet condition; the *Riccati diffusion* characterising its ground state $\Lambda_0(\beta)$ and the identity $-\Lambda_0(\beta) \stackrel{d}{=} \mathrm{TW}_\beta$. Note the dictionary $\eta = 2/\sqrt{\beta}$, i.e. $\beta = 4/\eta^2$ — the paper's inner equation is exactly RRV's diffusion.
- KPZ / edge universality context: why the Airy/Tracy–Widom object is universal.

**Primary reading.** Romik, *The Surprising Mathematics of Longest Increasing Subsequences* [free PDF online]— a gentle, complete route to Tracy–Widom. Anderson, Guionnet & Zeitouni, *An Introduction to Random Matrices* [free PDF online]— semicircle, edge, Airy kernel, TW, rigorously.

**Going deeper.** Ramírez, Rider & Virág, *Beta ensembles, stochastic Airy spectrum, and a diffusion*, JAMS (2011), arXiv:math/0607331 (ref. [3]) — read for the operator and the Riccati-diffusion characterisation. Tracy & Widom (1994), *Level-spacing distributions and the Airy kernel* (ref. [6]). Forrester, *Log-Gases and Random Matrices*. Corwin, *The KPZ equation and universality class* (ref. [10]).

**Milestone problems.**

1. From RRV, write the Riccati diffusion characterising $\Lambda_0(\beta)$ and identify it term-for-term with the swept inner Riccati (1) under $\eta = 2/\sqrt{\beta}$.
2. Show $-\Lambda_0(\infty) = a_1$ recovers the deterministic Airy-zero escape (the $\beta \to \infty$ limit).
3. Explain why $\beta = 1, 2, 4$ correspond to GOE/GUE/GSE and how the paper realises non-integer $\beta = 4/\eta^2$.

**What this unlocks in the paper**

§3 (statement of the laws), §4.2 (the operator $\mathcal{H}_\beta$ and $-\Lambda_0(\beta) \stackrel{d}{=} \mathrm{TW}_\beta$), the random-matrix cross-check in §8 (Figure 2), and the universality discussion in §9.

# 6 Track D — Painlevé II and the exact distributions

## 6.1 Module D — The Painlevé II transcendent, Hastings–McLeod, and the constant $c$

**Goal**

Know the Painlevé II equation and the Hastings–McLeod solution that gives the exact TW densities; and understand why the rate constant $c = 5.4439\ldots$ is a Painlevé II action (and emphatically not $2\pi$).

*Prerequisites:* F4, C.

**Core topics.**

- The six Painlevé equations and the Painlevé property; Painlevé II $s'' = xs - 2s^3$ (and the $\pm$ cubic variants); the Hastings–McLeod connection-problem solution.
- The Tracy–Widom formula: $F_2$ as a Fredholm determinant and via the Hastings–McLeod Painlevé II solution; the exact $\mathrm{TW}_1, \mathrm{TW}_2$ densities used (parameter-free) in the paper's Figure 1.
- Action functionals and Pohozaev identities for the nonlinear-Airy ground state $-g'' + xg = g^3$; the substitution $g = \sqrt{2}\,s$ mapping it to defocusing Painlevé II, exhibiting $c = \int g'^2$ as a Painlevé II action.

**Primary reading.** NIST DLMF ch. 32 (Painlevé) [free PDF online]. Clarkson, *Painlevé equations — nonlinear special functions* (survey) [free PDF online].

**Going deeper.** Fokas, Its, Kapaev & Novokshenov, *Painlevé Transcendents: The Riemann–Hilbert Approach* (AMS, 2006). Hastings & McLeod (1980), the original boundary-value problem.

**Milestone problems.**

1. Show $g = \sqrt{2}\,s$ turns $-g'' + xg = g^3$ into $s'' = xs - 2s^3$ (Painlevé II).
2. Use two Pohozaev identities to reduce the action $\frac{1}{2} \min K[\psi]^2 / \int \psi^4$ to $\int g'^2$ and confirm numerically $c = 5.4439\ldots$; explain why a finite-$\eta$ fit overestimates $c$ (the spurious 6.22) and why the answer is not $2\pi$.
3. Reproduce the exact $\mathrm{TW}_2$ density from the Hastings–McLeod solution and overlay a Monte-Carlo histogram of $Y_{\mathrm{node}}$ (Figure 1).

**What this unlocks in the paper**

The exact densities used parameter-free in §8; Theorem 3's identification of $c$ as a Painlevé II action; and the closed-form TW machinery underlying §3.

# 7 Track E — Excitable systems and the phase channel

## 7.1 Module E — Excitability, SNIC, the Adler equation, and noisy first passage on the circle

**Goal**

Understand the second noise channel: why the destroying bifurcation matters, how a saddle-node-on-invariant-circle (SNIC) creates an Adler bottleneck, and how rescaling produces the canonical noisy saddle-node with its 2/3 exponent.

*Prerequisites:* F1, B1 (B2 helpful).

**Core topics.**

- Excitable systems and Hodgkin's excitability classes; the bifurcations by which spiking onsets/offsets (fold of cycles, SNIC, homoclinic); FitzHugh–Nagumo as the running example.
- The phase-response curve and the phase reduction of an oscillator; why a finite-period (fold-of-cycles) death gives regular phase diffusion $D_\varphi \sim \sigma^2$, but an infinite-period (SNIC) death is singular.
- The Adler equation $\mathrm{d}\theta = (1 - \cos\theta)\,\mathrm{d}t + \sigma\,\mathrm{d}W$; mean-first-passage-time quadrature on the tilted periodic potential; the rescaling $\theta = \sigma^{2/3}u$ to the parameter-free canonical noisy saddle-node $\mathrm{d}u = \frac{1}{2}u^2\,\mathrm{d}\tau + \mathrm{d}W$; Watson's lemma giving $\omega, D_\varphi \sim \sigma^{2/3}$ with rate constant $\pi/J$, $J = \iint_{w<u} e^{(w^3-u^3)/3}$.

**Primary reading.** Izhikevich, *Dynamical Systems in Neuroscience*, ch. 6–7 (excitability, SNIC) and ch. 10 (phase models). Ermentrout & Terman, *Mathematical Foundations of Neuroscience* (ref. [11]), ch. on oscillations and PRCs.

**Going deeper.** Reimann *et al.*, *Giant acceleration of free diffusion by use of tilted periodic potentials*, PRL (2001) (ref. [9]) — the MFPT quadrature. Strogatz ch. 4/8 for the deterministic Adler equation.

**Milestone problems.**

1. Nondimensionalise the Adler bottleneck $\mathrm{d}\theta = (1-\cos\theta)\mathrm{d}t + \sigma\,\mathrm{d}W$ to $\mathrm{d}u = \frac{1}{2}u^2\mathrm{d}\tau + \mathrm{d}W$ via $\theta = \sigma^{2/3}u$ near the degenerate saddle.
2. Use the MFPT quadrature and Watson's lemma to obtain $\omega = (\pi/J)\sigma^{2/3}(1 + o(1))$ and $D_\varphi \sim \sigma^{2/3}$.
3. Contrast with the fold-of-cycles case and show the phase diffusion is then regular, $D_\varphi \sim \sigma^2$.

**What this unlocks in the paper**

All of §7 (the fold-of-cycles vs. SNIC dichotomy, the canonical noisy saddle-node, the 2/3 exponents, the constant $J$) and the application discussion in §9.2.

# 8 Capstone — assembling the proof

## 8.1 Module S1 — The shooting characterisation (the keystone, §4.3)

**Goal**

Master the single hardest original step: that the law of the *swept* first node equals the *fixed* half-line operator's ground state. This is where Tracks A, B, C and module F3 meet.

*Prerequisites:* A, B1–B2, C, F3.

**What to work through.**

- Reread §4.1–4.3 with the companion note *ShootingCharacterisation proof*. Reconstruct: (i) define $G(\ell)$, the ground-state eigenvalue of $M = -\partial_Y^2 + Y - \eta\tilde{\xi}$ on $[\ell, \infty)$ with Dirichlet data at $\ell$; (ii) Dirichlet domain monotonicity makes $G$ strictly increasing, so the pathwise identity $\{Y_{\text{node}} \le t\} = \{G(t) \ge 0\}$ holds; (iii) the level $t$ is now *deterministic*, so the shift $Y = t + x$ and the *stationarity and reflection symmetry of white noise* give $G(t) \stackrel{d}{=} \Lambda_0(\beta) + t$; (iv) conclude $Y_{\text{node}} \stackrel{d}{=} -\Lambda_0(\beta) \stackrel{d}{=} \mathrm{TW}_\beta$.
- Understand the "forgetting lemma": because the canard is attracting, the finite-$Y_0$ sweep converges to the recessive solution, linking $Y_{\text{node}}$ to the actual dynamical passage.
- Pin down precisely which input is the cited black box (the RRV meaning of $\mathcal{H}_\beta$) and which is proved here (the monotone spectral function removing the random-level obstruction).

**What this unlocks in the paper**

Theorem 2 in full — the reduction and shooting characterisation, including the resolution of the "random level $\Rightarrow$ noise-dependent shift" obstruction.

## 8.2 Module S2 — Full read-through and reproduction

**Goal**

Read the paper end to end, mapping each result to the module that supports it, and (optionally) reproduce the numerics.

*Prerequisites:* all.

**Section-by-section dependency map.**

| Paper element | Supported by |
|---|---|
| §1 Introduction; eq. (1) inner Riccati | A, F4 |
| §2 Normal form, blow-up, charts; $\eta = \sigma/\sqrt{\varepsilon_2}$ | A (B1 for Prop. 8) |
| §2.3 Two channels; hazard (5) | B2, F4 |
| §3 Main theorems (statements) | C, D |
| §4.1 Cole–Hopf, Lemma 6 | F4, B1 |
| §4.2 Stochastic Airy operator; Sturm fact | C, F3 |
| §4.3 Shooting characterisation (Thm 2) | **S1** = A+B+C+F3 |
| §5 Escape probability and rate constant $c$ | B3, D |
| §6 $\theta$-uniformity, composition, blow-down (Prop. 8) | A, B1–B2 |
| §7 Phase channel; SNIC; 2/3 exponent | E, F4 |
| §8 Numerical validation | C, D (B for Monte-Carlo) |
| §9 Universality and applications | C (KPZ), E |

**Companion technical notes (named in the paper).** *ShootingCharacterisation*, *InstantonPrinciple*, *CompositionPropagation*, *BlowDown*, *ChannelB SNIC*, *ChannelB rigour*. Read each against the module that prepares it (S1; B3+D; B2; B1+A; E; E).

**Optional reproduction (consolidates everything).**

1. Monte-Carlo the inner Riccati (1) for $\beta = 1, 2$ and overlay the exact TW densities with no fitting (Figure 1).
2. Independently discretise $\mathcal{H}_\beta$ as a random tridiagonal matrix and compare its smallest eigenvalue's law to the ODE shooting result (Figure 2; Bornemann, ref. [7], for TW numerics).
3. Estimate the local slope $-\mathrm{d}\log P_{\mathrm{early}}/\mathrm{d}(\eta^{-2})$ and watch it descend toward $c = 5.444$ (Figure 3); confirm the early-escape tail values ($\beta = 2$: 0.0298 vs. 0.0306; $\beta = 1$: 0.1644 vs. 0.1681) and the $\eta$-collapse across a factor four in $\varepsilon_2$.

## 9 Consolidated annotated reading list

**Foundations.**

- R. Durrett, *Probability: Theory and Examples*, 5th ed., Cambridge, 2019. Measure-theoretic probability. [free PDF online] `https://services.math.duke.edu/~rtd/PTE/pte.html`
- D. Williams, *Probability with Martingales*, Cambridge, 1991. Clean martingale theory.
- G. Teschl, *Mathematical Methods in Quantum Mechanics*, AMS GSM 157. Functional analysis, self-adjointness, Sturm–Liouville and oscillation theory. [free PDF online] `https://www.mat.univie.ac.at/~gerald/ftp/book-schroe/schroe.pdf`
- C. Bender & S. Orszag, *Advanced Mathematical Methods for Scientists and Engineers*, Springer, 1999. WKB, asymptotics, Airy.
- NIST *Digital Library of Mathematical Functions*. Airy (ch. 9), Painlevé (ch. 32). [free PDF online] `https://dlmf.nist.gov/`
- Yu. Kuznetsov, *Elements of Applied Bifurcation Theory*, Springer. Rigorous normal forms, centre manifolds, periodic-orbit bifurcations.
- S. Strogatz, *Nonlinear Dynamics and Chaos*. Geometric refresher; Adler equation.

**Track A — GSPT and blow-up.**

- C. Kuehn, *Multiple Time Scale Dynamics*, Springer AMS 191, 2015. The core text: GSPT, blow-up, canards, oscillations.
- H. Jardón-Kojakhmetov & C. Kuehn, *A survey on the blow-up method for fast-slow systems*, arXiv:1901.01402. [free PDF online] `https://arxiv.org/abs/1901.01402`
- M. Krupa & P. Szmolyan, *Extending GSPT to nonhyperbolic points — fold and canard points*, SIAM J. Math. Anal. 33 (2001) 286–314. (ref. [1])
- M. Desroches *et al.*, *Mixed-Mode Oscillations with Multiple Time Scales*, SIAM Review 54 (2012) 211–288. [free PDF online] `https://hal.science/hal-00765216`
- S. Jelbart, C. Kuehn & S.-V. Kuntz, *Geometric Blow-Up for Folded Limit Cycle Manifolds in Three Time-Scale Systems*, J. Nonlinear Sci. 34 (2024); arXiv:2208.01361 (ref. [2], the deterministic backbone). `https://arxiv.org/abs/2208.01361`

**Track B — stochastic analysis and large deviations.**

- L. C. Evans, *An Introduction to Stochastic Differential Equations*, AMS. [free PDF online] `https://math.berkeley.edu/~evans/SDE.course.pdf`
- B. Øksendal, *Stochastic Differential Equations: An Introduction with Applications*, Springer.
- I. Karatzas & S. Shreve, *Brownian Motion and Stochastic Calculus*, Springer. Rigorous reference.

- N. Berglund & B. Gentz, *Noise-Induced Phenomena in Slow–Fast Dynamical Systems: A Sample-Paths Approach*, Springer, 2006. (ref. [4])
- N. Berglund, B. Gentz & C. Kuehn, *Hunting French ducks in a noisy environment*, JDE 252 (2012); *From random Poincaré maps to stochastic MMO patterns*, JDDE 27 (2015). (ref. [5])
- M. Freidlin & A. Wentzell, *Random Perturbations of Dynamical Systems*, Springer.
- A. Dembo & O. Zeitouni, *Large Deviations Techniques and Applications*, 2nd ed., Springer, 1998. (ref. [8])

**Track C — random matrices.**

- D. Romik, *The Surprising Mathematics of Longest Increasing Subsequences*, Cambridge, 2015. Gentle route to Tracy–Widom. [free PDF online] https://www.math.ucdavis.edu/~romik/surprising-mathematics-of-longest-increasing-subsequences/
- G. Anderson, A. Guionnet & O. Zeitouni, *An Introduction to Random Matrices*, Cambridge, 2010. [free PDF online] https://cims.nyu.edu/~zeitouni/cupbook.pdf
- J. Ramírez, B. Rider & B. Virág, *Beta ensembles, stochastic Airy spectrum, and a diffusion*, JAMS 24 (2011) 919–944; arXiv:math/0607331 (ref. [3]). https://arxiv.org/abs/math/0607331
- C. Tracy & H. Widom, *Level-spacing distributions and the Airy kernel*, Comm. Math. Phys. 159 (1994) 151–174. (ref. [6])
- P. Forrester, *Log-Gases and Random Matrices*, Princeton, 2010.
- I. Corwin, *The Kardar–Parisi–Zhang equation and universality class*, Random Matrices Theory Appl. 1 (2012). (ref. [10])
- F. Bornemann, *On the numerical evaluation of distributions in random matrix theory*, Markov Process. Related Fields 16 (2010). (ref. [7])

**Track D — Painlevé.**

- P. Clarkson, *Painlevé equations — nonlinear special functions* (survey/lecture notes). [free PDF online]
- A. Fokas, A. Its, A. Kapaev & V. Novokshenov, *Painlevé Transcendents: The Riemann–Hilbert Approach*, AMS Surveys & Monographs 128, 2006.
- S. Hastings & J. McLeod, *A boundary value problem associated with the second Painlevé transcendent...*, Arch. Rational Mech. Anal. (1980).

**Track E — excitable systems.**

- E. Izhikevich, *Dynamical Systems in Neuroscience: The Geometry of Excitability and Bursting*, MIT Press, 2007.
- G. B. Ermentrout & D. Terman, *Mathematical Foundations of Neuroscience*, Springer, 2010. (ref. [11])
- P. Reimann *et al.*, *Giant acceleration of free diffusion by use of tilted periodic potentials*, Phys. Rev. Lett. 87 (2001) 010602. (ref. [9])

## 10 Milestone checklist

| Module | You can do this | Done |
|---|---|---|
| F1 | Derive the saddle-node normal form; recognise a fold of cycles in a Poincaré map | ☐ |
| F2 | State "$\stackrel{d}{=}$" precisely; prove BM stationarity/reflection | ☐ |
| F3 | Prove Dirichlet domain monotonicity and the Sturm first-zero = ground-state fact | ☐ |
| F4 | Show $\mathrm{Ai}'/\mathrm{Ai}$ solves the Riccati and escapes at $a_1$; do the Cole–Hopf algebra | ☐ |
| A | Run the $K_2$ rescaling to get (1) with $\eta = \sigma/\sqrt{\varepsilon_2}$ | ☐ |
| B1 | Prove Lemma 6 (no Itô correction) and Proposition 8 | ☐ |
| B2 | Reproduce the hazard (5); set up the telescoping error bound | ☐ |
| B3 | Run Schilder + contraction to the variational problem for $c$ | ☐ |
| C | Match the RRV Riccati diffusion to (1); explain $-\Lambda_0(\beta) \stackrel{d}{=} \mathrm{TW}_\beta$ | ☐ |
| D | Map $-g'' + xg = g^3$ to Painlevé II; reduce $c$ to $\int g'^2 = 5.4439\ldots$ | ☐ |
| E | Rescale the Adler bottleneck to $\mathrm{d}u = \frac{1}{2}u^2\mathrm{d}\tau + \mathrm{d}W$; get the 2/3 law | ☐ |
| S1 | Reconstruct the shooting characterisation end to end | ☐ |
| S2 | Read the paper, mapping every result to its module; reproduce a figure | ☐ |

## 11 "Can you explain it?" — the final self-test

You can claim to understand and explain the paper when you can answer each of these in a couple of minutes, at the board, without notes.

1. What is a *folded limit cycle*, and how does it differ from a fold point and a folded node?
2. Why does slow passage through a fold create a *canard*, and what is the deterministic escape point ($a_1 \approx -2.3381$)?
3. What does the geometric blow-up accomplish, and why are the weights $(1, 2, 3)$ the natural ones?
4. State the inner Riccati equation (1) and explain the effective noise $\eta = \sigma/\sqrt{\varepsilon_2}$.
5. Why does the additive noise need *no* Itô–Stratonovich correction (Lemma 6)?
6. What does the Cole–Hopf substitution $R = -u'/u$ achieve, and why is a blow-up of $R$ exactly a node of $u$?
7. What is the stochastic Airy operator $\mathcal{H}_\beta$, and why is its ground state Tracy–Widom distributed?
8. State the dictionary $\beta = 4/\eta^2$ and Theorem 2 in words: the peel-off level is $\mathrm{TW}_\beta$.

9. Why is the swept-first-node law equal to the fixed half-line operator's ground state — i.e. what does the monotone spectral function $G$ fix (the keystone, §4.3)?
10. Where does the early-escape probability $1 - F_\beta(0)$ come from, and how does it differ from the hazard $H = \eta^2/4\pi$?
11. Why is the rate constant $c = 5.4439\ldots$ a Painlevé II action and not $2\pi$?
12. In the phase channel, why does SNIC (infinite period) give the $\sigma^{2/3}$ law while a fold of cycles (finite period) gives regular $\sigma^2$ diffusion?
13. State the universality claim (Tracy–Widom / KPZ edge class) and its concrete physical predictions: $r, y \sim (\sigma^{2/3}, \sigma^{4/3})$ and $\sigma_* \propto \sqrt{\varepsilon_2}$.